

Multiple Choice

1. **Answer:** A

Options: A) High detrital biomass; B) High number of predatory species; C) High presence of fully matured plant species; D) High frequency of R-selected species; E) High rates of soil nutrient depletion

Note: Correct option: A - High detrital biomass

2. **Answer:** E

Options: A) It increases the acid concentration in the mitochondria matrix.; B) It decreases the acid concentration in the mitochondria matrix.; C) It increases the concentration of H⁺ in the mitochondria matrix.; D) It decreases the OH⁻ concentration in the mitochondria matrix.; E) It increases diffusion of H⁺ from the intermembrane space to the matrix.

Note: Correct option: E - It increases diffusion of H⁺ from the intermembrane space to the matrix.

3. **Answer:** A

Options: A) the tRNA transcript from the original DNA; B) a single strand of the original DNA segment after a substitution mutation; C) the final processed mRNA made from the original DNA; D) the primary RNA transcript (after splicing) from the original DNA; E) a single strand of the original DNA segment after a duplication mutation

Note: Correct option: A - the tRNA transcript from the original DNA

4. **Answer:** A

Options: A) thrombin; B) glycosaminoglycans; C) collagens; D) fibroblasts

Note: Correct option: A - thrombin

5. **Answer:** A

Options: A) Water is useful just because of its cooling properties.; B) Water is only important for hydration.; C) Water's role is limited to acting as a chemical reactant in cellular processes.; D) Water is important primarily because it contributes to the structural rigidity of cells.; E) Water's only function is to provide a medium for aquatic organisms.

Note: Correct option: A - Water is useful just because of its cooling properties.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The neurilemma sheath provides a channel for the axon to grow back to its former position.'

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'causes cell death'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Water moves from the surrounding water into the potato.'

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Without HCl, more bacteria would be killed in the stomach.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** The parathyroid glands produce insulin to regulate blood sugar levels. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key